

FPAS Mark II: The Least-Regrets Mindset

Note

1. Historical Lineage

The Reserve Bank of New Zealand launched the first inflation-targeting regime in 1990 and, by 1998, published a comprehensive manual outlining the Forecasting and Policy Analysis System (FPAS). This framework set the international standard for disciplined, forward-looking monetary policy. Many central banks - including Canada, Chile, the Czech Republic, and Israel - adopted the FPAS principles, integrating them into their inflation-targeting systems.

But over time, several large, advanced-economy institutions - notably the Federal Reserve, the Bank of England, the European Central Bank, the Bank of Japan, and the Bank of Korea failed to develop comparable systems. They remained reliant on fragmented forecasting methods and judgmental processes without a coherent analytical framework linking models, scenarios, and policy decisions.

After the global financial crisis, the RBNZ began using the term “least regrets” to describe a cautious approach to balancing risks between over-tightening and under-tightening policy. However, this remained a communications device, not a transformation of the policy framework itself. No changes were made to the structure of its FPAS or to the way risks were managed and formally integrated into the decision process.

2. The Analytical Leap

The Central Bank of Armenia’s innovation was to take the “least regrets” concept and operationalize it through FPAS Mark II. Mark II is designed for a world characterized by deep uncertainty - where the nature, persistence, and interaction of shocks are impossible to model into one optimal baseline projection.

Instead of optimizing policy around a single baseline scenario, Mark II embeds risk analysis into every stage of the policy process. Uncertainty is not treated as noise but as the core feature of the environment in which policy must operate.

Mark II combines several key elements:

1. Modelling framework capable of handling nonlinearities in the economy and endogenous policy credibility.
2. A decision framework based on management and least-regrets analysis.

3. A communication strategy emphasizing transparency about possible policy responses under uncertainty rather than false precision.
4. Continuous development of human capital in a dynamic learning environment.

This integration transforms monetary policy from a static optimization problem into a dynamic process of risk management and learning.

3. Institutional Comparisons

The difference between Armenia and other central banks is not one of resources but of institutional mindset. Large institutions - such as the Fed, ECB, and Bank of England - continue to operate without integrating analysis, decision-making, and communication. They have not yet built even the first generation of FPAS that New Zealand pioneered in the 1990s.

In contrast, Armenia developed the second generation - FPAS Mark II - by combining structured projection and policy analysis with the flexibility of prudent risk management. This was a leap forward in institutional design, not an incremental upgrade.

Waller's observation that "Armenia has become the new New Zealand" reflects the growing recognition that innovation in monetary frameworks increasingly comes from smaller, agile institutions.

4. The Least-Regrets Principle in Practice

In FPAS Mark II, every policy round begins with a structured scenario matrix. Staff with close and continuous involvement of Board members jointly evaluates risks and uncertainties in the economy. At the beginning of each policymaking round, Board members independently prepare one-page initial macroeconomic risk assessments, where they list out the main risks and uncertainties (subject to the 3R constraint) that they deem to be the most pressing and worthy of internal discussion and external communication. The staff's role in the meantime is, based on deliberations with Board members, preparing high-quality and reasonable risk scenarios.

In FPAS Mark II framework, at least 3 types of scenarios are prepared: a scenario that is based on market's expectations, scenarios that result in the policy rate following a higher path than what is priced in markets (Case A) and scenarios where the policy rate would follow a lower path than what is currently priced into markets (Case B). Additionally, when appropriate, tail risk scenarios including Case X (policy rate path significantly higher than market expectations) and Case Y (significantly below market expectations) could be included, which would need to be managed to avoid throwing the economy into dark corners.

For each scenario, the analysis evaluates the costs of acting too soon versus too late. The objective is to choose the policy stance that minimizes potential regret across all possible futures.

Mistakes are asymmetric:

- Tightening too early may unnecessarily suppress demand and employment.
- Tightening too late may allow inflation expectations to de-anchor.

The art of policymaking under FPAS Mark II is selecting the stance that minimises policy error across all plausible outcomes - not optimizing for a single point forecast.

5. Implications for Credibility and Communication

The least-regrets framework transforms communication. Instead of presenting a single baseline projection, the central bank explains:

- how it manages uncertainty,
- how it evaluates risks,
- how it would respond if conditions differed from expectations.

This shift makes transparency meaningful: the public understands both what the central bank expects and how it plans to act across different environments e.g. if certain risks materialize or economic conditions change.

FPAS Mark II completes the intellectual journey that began in New Zealand after the global financial crisis. For the first time, the least-regrets idea has moved from language to institutional reality. Christopher Waller, while at a conference in June 2025 at the Bank of Korea, recognized this achievement when he remarked that once he understood what had been built in Armenia, he realized that Armenia had “become the new New Zealand.” This statement captured a generational shift — from New Zealand as the pioneer of the first FPAS in 1990 to Armenia as the pioneer of the second generation, FPAS Mark II.

Literature & Further Reading

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Script

Hello everyone, I’m Susanna Grigoryan, an economist at the Central Bank of Armenia and a Level One student at the Global Forecasting School. In this video, we’ll explore FPAS Mark II — the least-regrets mindset — and how it represents a decisive intellectual leap in modern monetary policy.

The evolution from FPAS Mark I to FPAS Mark II represents a decisive intellectual leap in modern monetary policy. The term “least regrets” was first popularized by the Reserve Bank of New Zealand after the global financial crisis, but it remained a rhetorical concept rather than an operational framework. In contrast, the Central Bank of Armenia transformed the idea into a full analytical system. Christopher Waller, while at the “Structural Shifts and Monetary Policy” conference held at Bank of Korea from June 1-3 recognized this achievement when he remarked that once he understood what had been built in Armenia, he realized that Armenia had “become the new New Zealand.” This statement captured a generational shift — from New Zealand as the pioneer of the first FPAS in 1990 to Armenia as the pioneer of the second generation, FPAS Mark II.

The Reserve Bank of New Zealand launched the first inflation-targeting regime in 1990 and, by 1998, published a comprehensive manual outlining the Forecasting and Policy Analysis System, or FPAS. This framework set the international standard for disciplined, forward-looking monetary policy. Many central banks, including Canada, Chile, the Czech Republic, and Israel, adopted the FPAS principles, integrating them into their inflation-targeting systems. But as time passed, several large, advanced-economy institutions — notably the Federal Reserve, the Bank of England, the European Central Bank, the Bank of Japan, and the Bank of Korea — failed to develop comparable systems. They remained reliant on fragmented forecasting methods and judgmental processes without a coherent analytical framework linking models, scenarios, and policy decisions.

After the global financial crisis, the Reserve Bank of New Zealand began using the term “least regrets” to describe its cautious approach to balancing risks between over-tightening and under-tightening policy. However, this remained a communications device, not a transformation of the policy framework itself. No changes were made to the structure of its FPAS or to the way risks were formally integrated into the decision process.

The Central Bank of Armenia’s innovation was to take the “least regrets” concept and operationalize it through FPAS Mark II. Mark II is designed for a world characterized by deep uncertainty — where the nature, persistence, and interaction of shocks are impossible to impossible to model into one optimal baseline projection. Instead of optimizing policy around a single baseline scenario, Mark II embeds risk analysis into every stage of the policy process. This means that uncertainty is not treated as noise but as the core feature of the environment in which policy must operate.

The analytical foundation of Mark II combines several key elements. First, a modelling framework capable of handling nonlinearities in the economy and endogenous policy credibility. Second, a decision framework based on risk management and least-regrets analysis. Third, a communication strategy that emphasizes transparency about

uncertainty rather than false precision. Forth, realization that human capital is the key and need to be continuously developed in a dynamic learning environment. This integration transforms monetary policy from a static optimization problem into a dynamic process of risk management and learning.

The difference between Armenia and other central banks is not one of resources but of institutional mindset. Large, established institutions such as the Fed, the ECB, and the Bank of England continue to operate without fully integrating analysis, decision-making, and communication. They have yet to build even the first generation of FPAS that New Zealand pioneered in the 1990s. In contrast, Armenia has already developed the second generation directly — FPAS Mark II — by combining the discipline of structured forecasting with the flexibility of prudent risk management. This was a leap forward in institutional design, not an incremental upgrade.

Waller's observation that "Armenia has become the new New Zealand" is significant because it reflects the recognition that innovation in monetary policy now comes from smaller, more agile institutions. Just as New Zealand redefined central banking three decades ago by introducing inflation targeting, Armenia is now redefining it again by introducing a full risk-management-based policy framework.

In FPAS Mark II, every policy round begins with a structured scenario matrix. Staff with close and continuous involvement of Board members jointly evaluates risks and uncertainties in the economy. At the beginning of each policymaking round, Board members independently prepare one-page initial macroeconomic risk assessments, where they list out the main risks and uncertainties (subject to the 3R constraint) that they deem to be the most pressing and worthy of internal discussion and external communication. The staff's role in the meantime is, based on deliberations with Board members, preparing high-quality and reasonable risk scenarios.

In FPAS Mark II framework, at least 3 types of scenarios are prepared: a scenario that is based on market's expectations, scenarios that result in the policy rate following a higher path than what is priced in markets (Case A) and scenarios where the policy rate would follow a lower path than what is currently priced into markets (Case B). Additionally, when appropriate, tail risk scenarios including Case X (policy rate path significantly higher than market expectations) and Case Y (significantly below market expectations) could be included, which would need to be managed to avoid throwing the economy into dark corners.

The least-regrets mindset recognizes that policy mistakes are asymmetric. For example, tightening too early could unnecessarily suppress demand and employment, while

tightening too late could allow inflation expectations to de-anchor. The art of policymaking under FPAS Mark II is to find the policy stance that performs acceptably across all plausible outcomes, not to optimize one point forecast. This approach provides stability and resilience, even when the economy is hit by multiple interacting shocks.

The least-regrets framework also transforms communication. Instead of presenting a single baseline projection, the central bank explains how it manages uncertainty and evaluates trade-offs. This makes transparency meaningful: the public sees not only what the central bank expects, but how it would respond if the world evolves differently. In a world where credibility depends on humility and adaptability, this approach represents a major advance in communication strategy.

FPAS Mark II thus completes the intellectual journey that began in New Zealand in 1990. It integrates the lessons of three decades of inflation targeting and provides the blueprint for the next generation of credible monetary policy frameworks. For the first time, the least-regrets idea has moved from language to institutional reality.

When the Reserve Bank of New Zealand pioneered inflation targeting in 1990, it changed the way the world thought about monetary policy. By 1998, it had published a full manual describing how a Forecasting and Policy Analysis System — or FPAS — should work. That became the blueprint for what we now call FPAS Mark I. For many years, this system worked well in both advanced and emerging economies. But over time, most of the large central banks — the Fed, the Bank of England, the ECB, the Bank of Japan, and the Bank of Korea — never developed their own versions of this framework. They relied on judgment and fragmented models rather than a coherent system for linking analysis, decisions, and communication.

After the global financial crisis, New Zealand began using the phrase “least regrets” to describe a cautious approach to policy. It was a good phrase, but nothing changed in the underlying framework. There was still only one baseline forecast, and uncertainty was treated as something outside the model.

Here in Armenia, we took that same idea and made it real. FPAS Mark II is the first complete system that turns least-regrets thinking into a disciplined analytical framework. It accepts that uncertainty is not temporary — it’s the permanent condition of policymaking. Instead of trying to predict a single future, we prepare for several plausible ones. We evaluate the costs of acting too soon or too late and then choose the policy that we would least regret if the world turns out differently than we expect.

This least-regrets mindset changes everything. It means we build our forecasts, our policy discussions, and our communication around uncertainty — not around false precision.

The result is a system that combines independence, accountability, transparency, and analytical rigor in a single process. The central bank communicates the risks it sees in the economy as well as the scenarios to deal with them. It's not about predicting the future perfectly, but about managing risk responsibly and having least regrets by minimizing policy error

So, when people say Armenia has become the new New Zealand, it's not just a compliment — it's a recognition that the frontier of central banking has moved. Independence gives us the authority to act. Prudence and transparency give us the legitimacy to lead.

Thank you for watching, and I would like to acknowledge **Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervolidze** at the **Global Forecasting School** for their guidance and support in producing this video.

See you in the next video!